Vincent Savignac

Curriculum Vitae

Ph.D. Candidate
Department of Astronomy & Astrophysics
University of California San Diego
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EDUCATION

Doctorate of Philosophy - Astronomy & Astrophysics
2024-
University of California San Diego
Department of Astronomy & Astrophysics
La Jolla, California

Master of Science (transferred to above) – Physics
2023-2024
McGill University
Department of Physics
Montreal, Quebec

Bachelor of Science – Honours Mathematics and Physics
2020-2023
McGill University
Department of Physics and Department of Mathematics and Statistics
Montreal, Quebec

Diplôme d'études collégiales (DÉC) – Natural Sciences, Pure and Applied Sciences Profile
2018-2020
Cégep de Saint-Jean-sur-Richelieu
Natural Sciences
Saint-Jean-sur-Richelieu, Quebec

PUBLICATIONS

Savignac, V. and Lee, E. J. 2026, "Constructing the Earth's Formation History Using Deep Mantle Noble Gas Reservoirs", The Planetary Science Journal, 7, 135. DOI

Vlahos, E., Chachan Y., Savignac, V. and Lee, E. J. 2024, "Impacting Atmospheres: How Late-stage Pollution Alters Exoplanet Composition", The Astrophysical Journal, 976, 237. DOI

Savignac, V. and Lee, E. J. 2024, "The Not-So Dramatic Effect of Advective Flows on Gas Accretion", The Astrophysical Journal, 973(2), 85. DOI

MANUSCRIPTS IN PREPARATION

Savignac, V. and Lee, E. J. 2026, "The Magnetic Life and Death of Ultra-Short-Period Planets"

RESEARCH EXPERIENCE

Graduate student researcher
2024-
Department of Astronomy & Astrophysics, University of California San Diego
Advisor: Prof. Eve J. Lee

- Modeling the magnetic interactions between low-mass ultra-short-period planets and their host star.

Graduate student researcher
2023-2024
Trottier Space Institute at McGill (TSI) and Institute for Research on Exoplanets (IREx)
Advisor: Prof. Eve J. Lee

- Constraining the formation conditions of the early Earth's atmosphere
- Developing mass predictions concerning the distribution of low-mass planets at large orbital distance

Undergraduate student researcher
2022-2023
Trottier Space Institute at McGill (TSI) and Institute for Research on Exoplanets (IREx)
Advisor: Prof. Eve J. Lee

- Numerical simulations of the formation of the gaseous envelope of rocky planets

- Effect of atmospheric recycling on planetary envelope growth

Undergraduate student researcher
2021
Department of Mathematics and Statistics, McGill University
Advisor: Dr. Elliot Paquette

- Connection between the architecture of neural networks and random matrix theory predictions
- Analysis of gradient–based optimization methods on randomized problems

RESEARCH PRESENTATIONS

Rocky Worlds 4 (Poster)
January 19-23 2026
Energy Academy Europe, Zernike Campus
Groningen, Netherlands

Know The Star Know Thy Planet 2 (Contributed talk)
February 2-7 2025
Caltech
Pasadena, USA

Exoplanets V (Poster)
June 16-21 2024
Exoplanets Conference Series
Leiden, Netherlands

CRAQ annual meeting (Talk)
May 7-9 2024
Center for Research in Astrophysics of Quebec (CRAQ)
Lac-à-l'Eau-Claire, Canada

Physics/TSI Summer Undergrad Research Showcase (Talk)
August 18 2023
Department of Physics, McGill University
Montreal, Canada

IREx Summer Interns' Final Presentations (Talk)
August 17 2023
Institute for Research on Exoplanets (IREx)
Montreal, Canada

CRAQ annual meeting (Talk)
May 9-11 2023
Center for Research in Astrophysics of Quebec (CRAQ)
Lac-à-l'Eau-Claire, Canada

Undergraduate Poster Showcase (Poster)
March 15 2023
McGill University
Montreal, Canada

Physics/MSI Summer Undergrad Research Showcase (Talk)
August 19 2022
Department of Physics, McGill University
Montreal, Canada

IREx Summer Interns' Final Presentations (Talk)
August 24 2022
Institute for Research on Exoplanets (IREx)
Montreal, Canada

4th Undergraduate Research Conference (Talk)
August 23 2021
Department of Mathematics and Statistics, McGill University
Montreal, Canada

AWARDS AND HONORS

NSERC Postgraduate Scholarships – Doctoral (PGS D)
2025-
Natural Sciences and Engineering Research Council of Canada (NSERC)

FRQNT Master research scholarships
2023-2025
Fonds de recherche du Québec – Nature et technologies (FRQNT)

NSERC Canada Graduate Scholarship-Master's (CGS M)
2023-2024
Natural Sciences and Engineering Research Council of Canada (NSERC)

Lorne Trottier Science Accelerator Fellowships
2023
Faculty of Science, McGill University

NSERC Undergraduate Student Research Award (USRA)
2023
Natural Sciences and Engineering Research Council of Canada (NSERC)

P.R. Wallace Prize in Physics
2023
McGill University

First Class Honours in Mathematics and Physics
2023
McGill University

Dean's Honour List
2023
McGill University

Physics/MSI Summer Undergrad Research Showcase - Honourable Mention
2022
Department of Physics, McGill University

Trottier Excellence Grant for Summer Interns
2022
Institute for Research on Exoplanets (IREx)

NSERC Undergraduate Student Research Award (USRA)
2022
Natural Sciences and Engineering Research Council of Canada (NSERC)

Garnet A. Woonton Prize in Physics
2021
McGill University

Dean's Honour List
2021
McGill University

McGill University Student Stipend
2021
McGill University

J. W. McConnell Scholarship
2020-2023
McGill University

One-Year Undergraduate Entrance Scholarship
2020
McGill University

TEACHING EXPERIENCE

Teaching Assistant: Astrophysical Fluid Dynamics 2026
Department of Astronomy & Astrophysics, University of California San Diego

Teaching Assistant: Galaxies and the Universe 2025
Department of Astronomy & Astrophysics, University of California San Diego

Teaching Assistant: Astrophysical Dynamics 2024
Department of Astronomy & Astrophysics, University of California San Diego

Teaching Assistant: Physics of Fluids 2024
Department of Physics, McGill University

Teaching Assistant: Introductory Physics - Mechanics 2023
Department of Physics, McGill University

Mathematics Tutor 2019-2022
Cégep de Saint-Jean-sur-Richelieu

OTHER EXPERIENCE

Participation in the Canadian Astroparticle Physics Summer School 2022
Arthur B. McDonald Canadian Astroparticle Physics Research Institute, Queen's University

- Introduction to the most recent advances in the search for dark matter and neutrino mass determination
- Laboratory training in Astroparticle Physics

Introductory stay at the Mont-Mégantic Observatory 2022
Mont-Mégantic Observatory (OMM)

- Introduction to observations in Astronomy
- Observation of exoplanet transits

SERVICE AND OUTREACH

Astronomy & Astrophysics student representative 2026
A&A-SIO joint seminar series

- Interdisciplinary seminar series that brings together the expertise in the Department of Astronomy & Astrophysics and the Scripps Institution of Oceanography at the University of California San Diego.

Cosmic tour participant 2026
UCSD Cosmic Tours

- Series of volunteer events offering portable planetarium shows to schools and outreach events in San Diego County.

Astronomy on Tap organizer 2026
Astronomy on Tap San Diego

- Series of free public lectures offered to the general audience in local bars.

McGill student representative for CRAQ 2024
Center for Research in Astrophysics of Quebec (CRAQ)

- Acting as the voice of my student body within a professional association of astrophysicists in Quebec.

"Invite an IREx Astronomer" program 2022
Institute for Research on Exoplanets (IREx)

- Interacting with a younger audience and introducing students to astronomy.

PROFESSIONAL SKILLS

Programming Languages
Python

Languages
English: fluent (speaking, reading, writing)
French: fluent (speaking, reading, writing)